

Q2: 2023 Backcast Revenue Calculation

GCV Solar + Storage Valuation — NYISO N.Y.C. Zone

E3 Applicant Exercise

2026

Inputs and Notation

All revenue is calculated from hourly 2023 NYISO N.Y.C. zone day-ahead prices. For hour t , let p_t be the day-ahead energy price (\$/MWh) and s_t be the normalized solar production shape. Solar output is assumed to be available for the full hour, so MW and MWh are numerically equivalent for each hourly interval.

Parameter	Value
Solar size	300 MW-DC
Solar inverter limit	250 MW-AC
Storage power capacity	50 MW
Storage energy capacity	200 MWh
Storage duration	200 MWh/50 MW = 4 hours
Round-trip efficiency	100% per provided input
Dispatch	Per-day LP optimum (multi-cycle), perfect day-ahead foresight

Note on price basis: we use day-ahead (DA) prices specifically because the storage dispatch model (Q2b/Q2c) assumes perfect foresight of which hours to charge and discharge within a day — an assumption that is realistic under DA prices, since they are actually published and known before the operating day, but would not be realistic under real-time (RT) prices, which are not knowable in advance. A parallel RT-basis run was also completed for comparison: it earns modestly more (solar \$18.91M, storage \$2.73M) because it implicitly credits the model with foreknowledge of prices an RT-only strategy would not actually have in advance — the DA-basis figures below are the more defensible base case.

Q2a. Standalone Solar Revenue

Hourly solar generation is calculated from the normalized production shape and capped at the inverter limit:

$$G_t = \min(s_t \times 300 \text{ MW}, 250 \text{ MW})$$

Annual backcast solar revenue is the sum of hourly delivered energy times the day-ahead price:

$$R_{\text{solar}} = \sum_{t=1}^{8760} G_t \times p_t$$

Using the 2023 hourly data:

$$\sum_t G_t = 551,454 \text{ MWh}$$

$$R_{\text{solar}} = \$18,803,819$$

$$\text{Solar capture price} = \frac{\$18,803,819}{551,454 \text{ MWh}} = \$34.10/\text{MWh}$$

The inverter cap clips 372 MWh, equal to 0.07% of gross solar generation, so clipping has a negligible effect on standalone solar revenue.

Q2b. Standalone Storage Revenue

Storage is modeled as daily energy arbitrage, but rather than assume a single fixed 4-hour charge / 4-hour discharge cycle, each day d is solved as its own linear program: choose hourly charge $c_h \geq 0$ and discharge $g_h \geq 0$ (both ≤ 50 MW) to

$$\max \sum_{h \in d} p_h (g_h - c_h)$$

subject to, for every hour h in day d :

$$0 \leq \sum_{h' \leq h, h' \in d} (c_{h'} - g_{h'}) \leq 200 \text{ MWh}$$

i.e. the running state of charge must stay within the battery’s 200 MWh energy limit at every hour, starting the day at zero, with 100% round-trip efficiency (per the given input, so no loss term). This lets the optimizer discover *multiple* charge/discharge cycles within a day when day-ahead prices oscillate enough to make it profitable, rather than assuming exactly one cycle.

Inspecting individual days confirms this is a real effect, not a solver artifact. For example, April 11 has two separate profitable round trips across the day (an overnight dip into a morning peak, and a midday dip into an evening peak) — the optimizer uses 8 charge hours and 8 discharge hours that day, double a single 4-hour/4-hour cycle, because day-ahead prices simply have more usable volatility in a day than one high/low pair can capture. Solving this per-day LP across all of 2023 earns 14.75% more than the naive “4 lowest-price hours charge, 4 highest-price hours discharge” heuristic (\$1.978M vs. \$1.724M) — large enough that we adopted the LP as the primary method rather than a footnote.

Annual standalone storage revenue is:

$$R_{\text{storage}} = \sum_{d=1}^{365} R_{\text{storage},d} = \$1,977,954$$

Normalized by installed capacity:

$$\frac{\$1,977,954}{50,000 \text{ kW}} = \$39.56/\text{kW-year}$$

$$\frac{\$1,977,954}{200,000 \text{ kWh}} = \$9.89/\text{kWh-year}$$

Caveat: this still assumes zero degradation/cycling cost, so additional cycling is “free” here — a real operator would cycle less than this LP optimum once a per-cycle degradation cost is included (Q3b). Treat this as an upper bound on what multi-cycle dispatch can add, not a realistic operating recommendation on its own.

Q2c. Combined Solar + Storage Revenue

The combined case keeps the same solar export calculation as standalone solar. Storage uses the same per-day LP as Q2b, except each hour’s charging can first draw up to $\min(C_t, 50 \text{ MW})$ for \$0 from that hour’s otherwise-clipped solar,

$$C_t = \max(s_t \times 300 \text{ MW} - 250 \text{ MW}, 0)$$

before any additional charging that hour is priced at the grid rate — the optimizer naturally prefers the \$0 option first since it strictly lowers cost. Discharge and the multi-cycle structure are otherwise identical to Q2b. Across 2023, the model uses:

$$\sum_t \min(C_t, 50) = 372 \text{ MWh (free)}, \quad \text{grid-charged} = 125,595 \text{ MWh}$$

The resulting combined revenues are:

$$R_{\text{combined solar}} = \$18,803,819$$

$$R_{\text{combined storage}} = \$1,987,978$$

$$R_{\text{combined total}} = \$18,803,819 + \$1,987,978 = \$20,791,797$$

The interaction benefit versus building standalone solar and standalone storage separately is:

$$R_{\text{combined total}} - (R_{\text{solar}} + R_{\text{storage}}) = \$20,791,797 - (\$18,803,819 + \$1,977,954) = \$10,023$$

$$\frac{\$10,023}{\$20,781,773} = 0.048\%$$

Q2d. Reserve (Headroom/Footroom) Scenario — Extension Beyond the Base Exercise

As a further scenario, we estimate additional revenue each configuration could earn by offering NYISO operating reserve capacity, at an assumed reserve price of \$1.25/MW-hr (midpoint of the \$1–1.5/MW range). *Headroom* is capacity available to increase output on call (up-reserve); *footroom* is capacity available to decrease output on call (down-reserve).

Solar cannot offer reserve at all. It is a variable, weather-dependent resource with no firm or dispatchable capacity commitment, so it does not qualify to provide NYISO reserve products — standalone solar earns \$0 in reserve revenue:

$$R_{\text{solar reserve}} = \$0$$

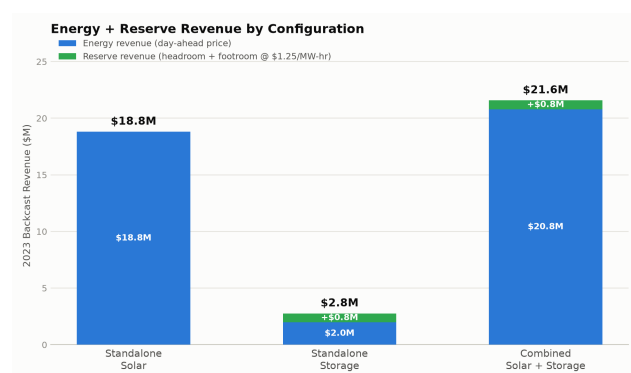
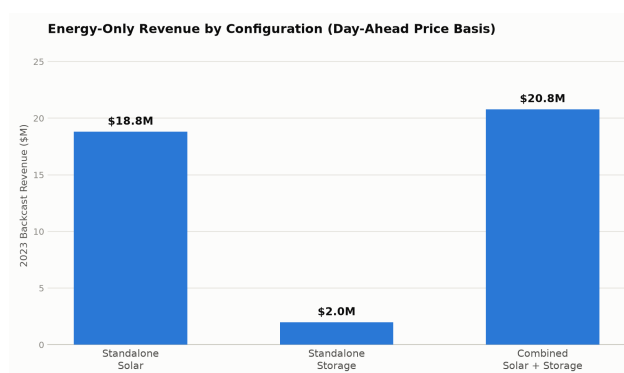
Storage’s reserve capacity is now classified hour-by-hour from the actual Q2b/Q2c LP dispatch solution, not a fixed 4-hour/4-hour/16-hour split: an hour where the LP charges (and doesn’t discharge) offers 50 MW of footroom only; an hour where it discharges (and doesn’t charge) offers 50 MW of headroom only; an idle hour offers the full 50 MW of *both* headroom and footroom simultaneously. Because multi-cycle dispatch (Q2b) uses more charge/discharge hours and leaves fewer hours idle than the old single-cycle heuristic, standalone storage actually has *less* total reserve capacity to offer here than it would under the simpler heuristic:

$$R_{\text{storage reserve}} = \$1.25 \times (312,550 + 312,550) \text{ MW-hr} = \$781,375$$

In the combined system, all reserve capacity comes from the storage component only — the solar component still cannot offer reserve — using the same hour-by-hour classification applied to the combined dispatch:

$$R_{\text{combined reserve}} = \$1.25 \times (312,550 + 312,950) \text{ MW-hr} = \$781,875$$

Configuration	Energy-Only	Reserve	Energy + Reserve	Reserve Uplift
Standalone solar	\$18.804M	\$0.000M	\$18.804M	0.0%
Standalone storage	\$1.978M	\$0.781M	\$2.759M	39.5%
Combined solar + storage	\$20.792M	\$0.782M	\$21.574M	3.8%



Storage is the only source of reserve value here (+39.5% on a standalone basis) because its reserve capacity comes free alongside energy dispatch, and idle hours can stack both headroom and footroom

at the same time. Solar contributes nothing, since it cannot commit firm capacity to a reserve product — so the combined system’s uplift (+3.8%) is diluted relative to standalone storage’s, simply because storage is a small share of the combined system’s total revenue. Note the direct tension with Q2b: multi-cycle dispatch earns more energy revenue precisely by keeping the battery busy charging or discharging more of the day, which leaves fewer idle hours available to sell as reserve — a real operator would face a genuine energy-vs-reserve tradeoff that this model does not co-optimize. This scenario is illustrative: real NYISO reserve products have stricter response-time, minimum-run, and co-optimization rules than modeled here, and a resource cannot literally be paid full price for headroom and footroom at the same instant if doing so double-counts the same physical capacity across products — a more rigorous treatment is one of the natural extensions noted in Q3b.

Summary

Configuration	2023 Backcast Revenue	Key Method
Standalone solar	\$18.804M	Hourly solar output $\times$ hourly DA price
Standalone storage	\$1.978M	Per-day LP-optimized multi-cycle arbitrage
Combined solar + storage	\$20.792M	Solar revenue plus storage with clipped solar charging first

The combined case produces only a small energy-arbitrage uplift over the standalone sum because the solar system clips very little energy in 2023. Figures above are energy-only; see Q2d for an illustrative reserve (headroom/footroom) scenario layered on top.